Revision Notes

CBSE Class 12 Chemistry • NCERT Part II, Chapters 6–10

Haloalkanes and Haloarenes • Alcohols, Phenols and Ethers • Aldehydes, Ketones and Carboxylic Acids • Amines • Biomolecules

How to use these notes. Keep the separate **reaction sheet** next to you. Each chapter has: mechanisms / derivations or key reactions; a **give-reasons bank** or conceptual questions; **exercise-wise solution patterns** (every NCERT exercise grouped into patterns, each with one solved question from the book, and a small line listing which questions belong to which pattern); exam tips; purple **JEE extras** (not needed for the board exam); and a **pattern spotter** for final revision. Question numbers match the NCERT textbook (Reprint 2026–27).

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Chapter 6 Haloalkanes and Haloarenes

6A Mechanisms the board asks

1. S_N2 (one step, inversion)

Nucleophile attacks from the back side as the halide leaves; transition state has 5 groups around C. Rate $= k[\text{RX}][\text{Nu}^-]$. Order: $CH_3X > 1^\circ > 2^\circ > 3^\circ$ (steric hindrance). Complete inversion (Walden).

2. S_N1 (two steps, racemisation)

Step 1 (slow): R–X $\rightarrow R^+ + X^-$. Step 2 (fast): $R^+ + Nu^-$. Rate $= k[\text{RX}]$. Order: $3^\circ > 2^\circ > 1^\circ > CH_3X$ (carbocation stability); benzylic/allylic fast. Polar protic solvents help. Optically active substrate gives a racemic mixture.

3. Elimination (Saytzeff rule)

Alcoholic KOH removes H from the β-carbon: the more substituted alkene is the major product. $CH_3CH_2CHBrCH_3 \xrightarrow{\text{alc. KOH}} CH_3CH{=}CHCH_3$ (major) + $CH_2{=}CHCH_2CH_3$.

6B Give-reasons bank

Reasoning questions

Statement	Reason
Haloarenes less reactive to nucleophilic substitution	Resonance gives C–X partial double bond; sp^2 C; phenyl cation unstable; repulsion from π electrons
$-NO_2$ at o/p increases reactivity of haloarenes	Withdraws electrons and stabilises the carbanion intermediate
Haloarenes give o/p products in electrophilic substitution, but slower than benzene	Halogen: $-I$ deactivates; $+R$ directs o/p
Grignard reagents need anhydrous conditions	They react with water to form alkanes
Alkyl halides, though polar, are immiscible with water	Cannot break H-bonds of water or form new ones
Dipole moment: chlorobenzene < cyclohexyl chloride	sp^2 C is more electronegative; resonance reduces C–Cl polarity
p-dichlorobenzene has higher m.p. than o/m	Symmetry: packs better in the crystal
Aqueous KOH gives alcohol, alcoholic KOH gives alkene	OH^- is a nucleophile in water; in alcohol RO^- acts as a strong base
KCN gives nitrile, AgCN gives isocyanide	KCN ionic: attack through C; AgCN covalent: attack through N (ambident CN^-)
Chloroform stored in dark bottles filled to the top	Oxidised by air in light to poisonous phosgene

$C_6H_5CHClC_6H_5$ hydrolyses faster than $C_6H_5CH_2Cl$	More stable carbocation (two phenyls), S_N1

6C Exercise-wise solution patterns

Exercises 6.1–6.22 (22 questions, 6 patterns)

Pattern A: Nomenclature, classification, isomers

Solved (Q6.1(i)): $(CH_3)_2CHCH(Cl)CH_3$: 2-chloro-3-methylbutane, secondary alkyl halide.
(Q6.6): C_4H_9Br: 1-bromobutane, 2-bromobutane, 1-bromo-2-methylpropane, 2-bromo-2-methylpropane.

Pattern B: Physical properties / identify a compound

Solved (Q6.4): CH_2Cl_2 has the highest dipole moment ($CCl_4 = 0$; in $CHCl_3$ the bond moments partly cancel).
(Q6.5): C_5H_{10} with no reaction in dark (not an alkene) and one monochloro product: cyclopentane.

Pattern C: S_N1/S_N2 reactivity order

Solved (Q6.16(i)): S_N2 order: 1-bromopentane > 2-bromopentane > 2-bromo-2-methylbutane (primary > secondary > tertiary).
(Q6.9): CH_3I faster than CH_3Br (I^- is a better leaving group); CH_3Cl faster than $(CH_3)_3CCl$.

Pattern D: Predict products (substitution vs elimination)

Solved (Q6.14): (ii) $(CH_3)_3CBr + KOH(alc) \rightarrow (CH_3)_2C{=}CH_2$; (iv) $CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN$; (v) $C_6H_5ONa + C_2H_5Cl \rightarrow C_6H_5OC_2H_5$; (viii) $CH_3CH{=}C(CH_3)_2 + HBr \rightarrow CH_3CH_2CBr(CH_3)_2$ (Markovnikov).
(Q6.10(ii)): 2-chloro-2-methylbutane: major 2-methylbut-2-ene, minor 2-methylbut-1-ene.

Pattern E: Conversions

Solved (Q6.19(viii)): aniline $\xrightarrow{NaNO_2/HCl,\ 273\ K}$ $C_6H_5N_2^+Cl^-$ $\xrightarrow{Cu_2Cl_2/HCl}$ chlorobenzene (Sandmeyer).
(Q6.19(xvi)): 2-bromopropane $\xrightarrow{\text{alc. KOH}}$ propene $\xrightarrow{HBr,\ \text{peroxide}}$ 1-bromopropane.

Pattern F: Word problems (identify A, B, C...)

Solved (Q6.21): (a) 1-bromo-2-methylpropane $\xrightarrow{\text{alc. KOH}}$ (b) 2-methylpropene $\xrightarrow{HBr}$ (c) 2-bromo-2-methylpropane (isomer of a).
(a) + Na (Wurtz) $\rightarrow$ (d) 2,5-dimethylhexane, different from *n*-octane given by *n*-butyl bromide.

Pattern A: Q6.1, 6.2, 6.3, 6.6. Pattern B: Q6.4, 6.5, 6.12, 6.13, 6.18. Pattern C: Q6.9, 6.16, 6.17. Pattern D: Q6.7, 6.8, 6.10, 6.14, 6.15, 6.20, 6.22. Pattern E: Q6.11, 6.19. Pattern F: Q6.21.

6D Exam tips

Common mistakes that lose marks

- Peroxide effect (anti-Markovnikov) works only with HBr.
- Wurtz needs *dry ether*; Finkelstein needs dry acetone (NaI); Swarts uses AgF/Hg_2F_2.
- Writing the minor alkene as the product of dehydrohalogenation.
- Forgetting stereochemistry words: inversion (S_N2), racemisation (S_N1).

Question types: 1-mark: IUPAC name, which reacts faster by S_N1/S_N2; 2–3 marks: named reactions, give reasons, conversions; 5 marks: mechanism, identify-the-compound problems.

6E JEE extras

Useful results

- Benzyne mechanism: chlorobenzene + $NaNH_2$ gives aniline (with ring-label scrambling).
- Nucleophilicity (polar aprotic): $I^- > Br^- > Cl^-$; leaving group ability I > Br > Cl > F.
- E2 needs anti-periplanar H and X; bulky base gives Hofmann (less substituted) alkene.

6F Pattern spotter

If you see...	Do this	See
Structure and properties		
Name / classify a halide	Longest chain, lowest locants; 1°/2°/3°, allyl, benzyl, vinyl, aryl	Ex A
Dipole moment, m.p., solubility	Symmetry, bond polarity, H-bonding	Ex B
Reactions		
Faster by S_N2	Less hindered, better leaving group	Ex C
Faster by S_N1	More stable carbocation	Ex C
Alcoholic KOH	Elimination, Saytzeff	Ex D
KCN / AgCN, KNO_2 / $AgNO_2$	Ambident: C vs N, O vs N attack	Ex D
A to B conversions	Chain ± 1 C, position change via alkene	Ex E
Mystery compound	Work forward from formula and tests	Ex F

Chapter 7 Alcohols, Phenols and Ethers

7A Mechanisms the board asks

1. Acid-catalysed hydration of ethene

(i) $H_2O + H^+ \longrightarrow H_3O^+$; ethene takes H^+ to give $CH_3CH_2^+$. (ii) H_2O attacks the carbocation: $CH_3CH_2OH_2^+$. (iii) loss of H^+ gives ethanol.

2. Dehydration of ethanol to ethene (443 K, conc. H_2SO_4)

(i) Protonation: $CH_3CH_2OH_2^+$. (ii) Loss of water (slow): $CH_3CH_2^+$. (iii) Loss of H^+: $CH_2{=}CH_2$. (At 413 K, ethanol attacks the protonated alcohol by S_N2 to give diethyl ether.)

3. Cleavage of ethers by HI

Protonation of ether O; I^- attacks the less hindered alkyl (S_N2): $CH_3OCH_3 + HI \longrightarrow CH_3I + CH_3OH$. With 3° alkyl: S_N1 gives 3° iodide. Anisole gives phenol + CH_3I (O–aryl bond does not break).

7B Give-reasons bank

Reasoning questions

Statement	Reason
Alcohols boil higher than hydrocarbons/ethers of similar mass	Intermolecular H-bonding
Lower alcohols are soluble in water	H-bonding with water; solubility falls as alkyl group grows
Phenol more acidic than ethanol	Phenoxide ion is resonance stabilised
o-nitrophenol more acidic than o-methoxyphenol	$-NO_2$ ($-I$, $-R$) stabilises phenoxide; $-OCH_3$ ($+R$) destabilises
o-nitrophenol is steam volatile	Intramolecular H-bonding; p-isomer has intermolecular H-bonding
$-OH$ activates benzene ring, o/p directing	$+R$ effect increases electron density at o and p
Ethers by acid dehydration of 2°/3° alcohols fail	Elimination to alkene competes and dominates
Williamson fails with 3° alkyl halides	Strong base causes elimination
Alkoxy group in anisole is o/p directing	$+R$ effect of O

7C Exercise-wise solution patterns

Exercises 7.1–7.33 (33 questions, 6 patterns)

Pattern A: Nomenclature and isomers

Solved (Q7.7): C_7H_8O phenols: 2-methylphenol, 3-methylphenol, 4-methylphenol (o-, m-, p-cresol).

Pattern B: Physical properties (H-bonding)

Solved (Q7.22): ethanol (b.p. 351 K) > methoxymethane: ethanol forms intermolecular H-bonds; the ether has no O–H.

Pattern C: Preparation reactions

Solved (Q7.9, cumene): $C_6H_5CH(CH_3)_2 \xrightarrow{O_2} C_6H_5C(CH_3)_2OOH \xrightarrow{H^+/H_2O} C_6H_5OH + CH_3COCH_3$.
(Q7.6, hydroboration): $CH_3CH{=}CH_2 \xrightarrow{(BH_3)_2} (CH_3CH_2CH_2)_3B \xrightarrow{H_2O_2/OH^-} CH_3CH_2CH_2OH$ (anti-Markovnikov).

Pattern D: Acidity and reactions of phenol

Solved (Q7.17): (ii) Br_2 in CS_2: o- and p-bromophenol; bromine water gives 2,4,6-tribromophenol. (iii) dilute HNO_3: o- and p-nitrophenol. (iv) $CHCl_3$/NaOH: salicylaldehyde (Reimer–Tiemann).

Pattern E: Conversions and reagents (Grignard, oxidation, reduction)

Solved (Q7.20(iv)): $CH_3MgBr + CH_3COCH_3 \longrightarrow (CH_3)_3COMgBr \xrightarrow{H_2O} (CH_3)_3COH$ (2-methylpropan-2-ol).
(Q7.21): 1° alcohol → acid: acidified $KMnO_4$; → aldehyde: PCC; ketone → 2° alcohol: $NaBH_4$.

Pattern F: Ethers (Williamson, cleavage, mechanisms)

Solved (Q7.24(iii)): 2-methoxy-2-methylpropane: use $(CH_3)_3CONa + CH_3Br$ (not $(CH_3)_3CBr + CH_3ONa$, which eliminates).
(Q7.28(ii)): anisole + HI → phenol + CH_3I.

Pattern A: Q7.1, 7.2, 7.3, 7.7, 7.23. Pattern B: Q7.4, 7.5, 7.8, 7.22. Pattern C: Q7.6, 7.9–7.13, 7.32. Pattern D: Q7.14–7.18. Pattern E: Q7.20, 7.21. Pattern F: Q7.11, 7.19, 7.24–7.31, 7.33.

7D Exam tips

Common mistakes that lose marks

- Lucas test: 3° turbid at once, 2° in 5 min, 1° not at room temperature.
- Phenol does not react with $NaHCO_3$ (carboxylic acids do).
- Kolbe: sodium phenoxide + CO_2 (400 K, 4–7 atm) then H^+ gives salicylic acid.
- In Williamson, choose the alkyl halide to be primary.

Question types: 1-mark: IUPAC names, reagent names; 2–3 marks: named reactions, acidity order, conversions; 5 marks: mechanisms (hydration, dehydration, HI cleavage).

7E JEE extras

Useful results

- Acidity of phenols: electron-withdrawing groups at o/p increase acidity: 2,4,6-trinitrophenol (picric acid) $\gg$ phenol $>$ cresol.
- Carbocation rearrangement (Q7.33): 2° cation shifts to 3° by hydride shift, giving 2-bromo-2-methylbutane.
- Pinacol–pinacolone rearrangement; oxidation with Cu at 573 K (1° $\rightarrow$ aldehyde, 2° $\rightarrow$ ketone, 3° $\rightarrow$ alkene).

7F Pattern spotter

If you see...	Do this	See
Properties		
Boiling point / solubility comparison	H-bonding	Ex B
Steam volatility	Intramolecular H-bond	Ex B
Acidity order of phenols	EWG increase, EDG decrease	Ex D
Reactions		
Alkene to alcohol	Hydration (Markovnikov) or hydroboration (anti)	Ex C
Phenol from benzene/cumene/chlorobenzene	Sulphonation + fusion; cumene; Dow	Ex C
Grignard + carbonyl	HCHO: 1°; RCHO: 2°; ketone: 3°	Ex E
Make an ether	Williamson with 1° halide	Ex F
Ether + HI	Smaller alkyl becomes iodide; aryl stays as phenol	Ex F

Chapter 8 Aldehydes, Ketones and Carboxylic Acids

8A Mechanisms the board asks

1. Nucleophilic addition to C=O

Nucleophile attacks the electrophilic carbonyl C (trigonal → tetrahedral alkoxide), then protonation. Reactivity: HCHO > RCHO > RCOR (steric + electronic $+I$ effects). Example: HCN gives cyanohydrin.

2. Aldol condensation (needs α-H)

$$2\,CH_3CHO \xrightarrow{\text{dil}\cdot\text{NaOH}} CH_3CH(OH)CH_2CHO \xrightarrow{\Delta} CH_3CH{=}CHCHO + H_2O.$$

3. Cannizzaro reaction (no α-H)

$2\,HCHO \xrightarrow{\text{conc}\cdot\text{KOH}} CH_3OH + HCOOK$; benzaldehyde gives benzyl alcohol + potassium benzoate.

8B Give-reasons bank

Reasoning questions

Statement	Reason
Aldehydes more reactive than ketones	Less steric hindrance; one alkyl ($+I$) group instead of two
Carboxylic acid stronger than phenol	Carboxylate has two equivalent resonance structures with negative charge on O; phenoxide puts charge on C
Cyclohexanone forms cyanohydrin well, 2,2,6-trimethylcyclohexanone does not	Steric hindrance by methyl groups
Only one NH_2 of semicarbazide reacts	The other is in resonance with C=O (amide-like), so not nucleophilic
Remove water or ester during esterification	Reversible reaction; shifts equilibrium forward
Carboxylic acids do not give carbonyl tests	Resonance reduces the C=O character
Carboxylic acids have high b.p.	Dimerisation by H-bonding
Electron-withdrawing groups increase acid strength	Stabilise the carboxylate ion

8C Exercise-wise solution patterns

Exercises 8.1–8.20 (20 questions, 7 patterns)

Pattern A: Nomenclature and derivatives

Solved (Q8.2(iv)): $CH_3COCH_2COCH_3$: pentane-2,4-dione. **(Q8.1(vi)):** oxime: $CH_3CHO + NH_2OH \longrightarrow CH_3CH{=}NOH + H_2O$.

Pattern B: Aldol vs Cannizzaro

Aldol needs α-H; Cannizzaro needs no α-H (aldehydes only). **Solved (Q8.7):** aldol: 2-methylpentanal, cyclohexanone, 1-phenylpropanone, phenylacetaldehyde. Cannizzaro: methanal, benzaldehyde, 2,2-dimethylbutanal. Neither: benzophenone (ketone without α-H), butan-1-ol (not a carbonyl).

Pattern C: Order of reactivity / acidity

Solved (Q8.12(i)): reactivity to HCN: di-tert-butyl ketone < methyl tert-butyl ketone < acetone < acetaldehyde.
(iii) acid strength: 4-methoxybenzoic < benzoic < 4-nitrobenzoic < 3,4-dinitrobenzoic acid.

Pattern D: Distinguishing tests

Solved (Q8.13): (i) propanal vs propanone: Tollens' silver mirror (aldehyde only). (ii) acetophenone vs benzophenone: iodoform test (CH_3CO- only). (iii) phenol vs benzoic acid: $NaHCO_3$ effervescence (acid) / $FeCl_3$ violet (phenol). (vii) ethanal vs propanal: iodoform (ethanal).

Pattern E: Predict products of reagents

Solved (Q8.6): cyclohexanecarbaldehyde + (i) $PhMgBr/H_3O^+$: cyclohexyl(phenyl)methanol; (ii) Tollens': cyclohexanecarboxylate; (iii) semicarbazide: semicarbazone; (iv) excess $EtOH/H^+$: acetal; (v) Zn–Hg/HCl: methylcyclohexane (Clemmensen).

Pattern F: Conversions

Solved (Q8.15(ii)): benzoic acid $\xrightarrow{SOCl_2}$ benzoyl chloride $\xrightarrow{H_2,\ Pd\text{–}BaSO_4}$ benzaldehyde (Rosenmund).
(Q8.15(i)): propanone $\xrightarrow{NaBH_4}$ propan-2-ol $\xrightarrow{\text{conc. } H_2SO_4,\ \Delta}$ propene.

Pattern G: Identify the compound

Solved (Q8.19): C:H:O $= \frac{69.77}{12} : \frac{11.63}{1} : \frac{18.6}{16} = 5 : 10 : 1$; M = 86: $C_5H_{10}O$. Not Tollens' (ketone), iodoform positive (CH_3CO-), oxidation to ethanoic + propanoic acids: **pentan-2-one**.
(Q8.10): $C_9H_{10}O$, aldehyde with Cannizzaro, oxidises to phthalic acid: 2-ethylbenzaldehyde.

Pattern A: Q8.1–8.5. Pattern B: Q8.7, 8.9. Pattern C: Q8.12, 8.20. Pattern D: Q8.13. Pattern E: Q8.6, 8.16, 8.17. Pattern F: Q8.8, 8.14, 8.15. Pattern G: Q8.10, 8.11, 8.19. Reasons: Q8.18.

8D Exam tips

Common mistakes that lose marks

- Iodoform test needs CH_3CO- or $CH_3CH(OH)-$ (ethanol, propan-2-ol also positive).
- Clemmensen (Zn–Hg/HCl) vs Wolff–Kishner (NH_2NH_2/KOH, glycol): both give CH_2.
- Fehling's: only aliphatic aldehydes (benzaldehyde negative); Tollens': all aldehydes.
- HVZ needs α-H in the acid.

Question types: 1-mark: name reactions, product of one step; 2–3 marks: tests, orders, conversions; 5 marks: identify a compound from data, mechanisms.

8E JEE extras

Useful results

- Crossed Cannizzaro: HCHO is always oxidised (to formate).
- Perkin, Claisen–Schmidt (benzaldehyde + acetone/acetophenone), Benzoin condensation.
- Acidity: $FCH_2COOH > ClCH_2COOH > BrCH_2COOH > CH_3COOH$; $HCOOH > CH_3COOH$; ortho effect: o-substituted benzoic acids are stronger.

8F Pattern spotter

If you see...	Do this	See
Reactivity		
Aldol or Cannizzaro?	α-H present: aldol; absent: Cannizzaro	Ex B
Order towards nucleophiles	Fewer/smaller alkyl groups react faster	Ex C
Acid strength order	EWG up, EDG down, distance matters	Ex C
Tests and products		
Aldehyde vs ketone	Tollens', Fehling's	Ex D
Methyl ketone	Iodoform	Ex D
Carboxylic acid	$NaHCO_3$ effervescence	Ex D
Reagent on a carbonyl	Grignard, Clemmensen, Wolff–Kishner, $NaBH_4$	Ex E
Acid to aldehyde	Acid chloride then Rosenmund	Ex F
Mystery compound	Empirical formula, then tests	Ex G

Chapter 9 Amines

9A Key reactions the board asks

1. Hoffmann bromamide degradation (one carbon less)

$\mathrm{RCONH_2 + Br_2 + 4\,NaOH \longrightarrow RNH_2 + Na_2CO_3 + 2\,NaBr + 2\,H_2O}$.

2. Gabriel phthalimide synthesis (pure 1° aliphatic amines)

Phthalimide $\xrightarrow{\mathrm{KOH}}$ potassium phthalimide $\xrightarrow{\mathrm{RX}}$ N-alkylphthalimide $\xrightarrow{\mathrm{NaOH(aq)}}$ $\mathrm{RNH_2}$. Not for aryl amines (aryl halides do not undergo $\mathrm{S_N}$ with phthalimide anion).

3. Diazotisation and diazonium reactions

$\mathrm{C_6H_5NH_2 + NaNO_2 + 2\,HCl \xrightarrow{273\text{-}278\,K} C_6H_5N_2^+Cl^- + NaCl + 2\,H_2O}$.
Replacement: $\mathrm{Cu_2Cl_2/HCl}$ → Cl; $\mathrm{Cu_2Br_2/HBr}$ → Br; CuCN → CN (Sandmeyer); KI → I; $\mathrm{HBF_4}$, heat → F; $\mathrm{H_2O}$ → OH; $\mathrm{H_3PO_2}$ or $\mathrm{C_2H_5OH}$ → H.
Coupling: with phenol (alkaline) gives orange p-hydroxyazobenzene; with aniline (acidic) yellow p-aminoazobenzene.

9B Give-reasons bank

Reasoning questions

Statement	Reason
$\mathrm{pK_b}$ of aniline higher than methylamine	Lone pair delocalised into the ring; less available
Ethylamine soluble in water, aniline not	H-bonding outweighed by the large hydrophobic phenyl group
Methylamine with $\mathrm{FeCl_3}$ gives $\mathrm{Fe(OH)_3}$ precipitate	Basic: produces $\mathrm{OH^-}$ in water
Aniline on nitration gives much m-nitroaniline	In acid, anilinium ion forms ($-\mathrm{NH_3^+}$ is m-directing)
Aniline does not undergo Friedel–Crafts	Forms a salt with $\mathrm{AlCl_3}$ (Lewis acid); N becomes positive and deactivates
Aryl diazonium salts more stable than alkyl	Resonance with the ring
Order of basicity in water: $\mathrm{(C_2H_5)_2NH} > \mathrm{(C_2H_5)_3N} > \mathrm{C_2H_5NH_2} > \mathrm{NH_3}$	Balance of $+I$ effect, solvation (H-bonding) and steric hindrance
1° amines boil higher than 3°	1° have two N–H for H-bonding; 3° have none
Amines less acidic than alcohols	N is less electronegative than O

9C Exercise-wise solution patterns

Exercises 9.1–9.14 (14 questions, 6 patterns)

Pattern A: Nomenclature and classification

Solved (Q9.1(iii)): $CH_3NHCH(CH_3)_2$: N-methylpropan-2-amine, secondary.

Pattern B: Distinguishing tests

Solved (Q9.2): (i) methylamine vs dimethylamine: carbylamine test (foul-smelling isocyanide only with 1°). (iii) ethylamine vs aniline: azo-dye test (aniline after diazotisation + β-naphthol gives orange dye). (ii) 2° vs 3°: Hinsberg (2° gives insoluble sulphonamide; 3° no reaction).

Pattern C: Basicity and physical property orders

Solved (Q9.4(iii)(a)): p-nitroaniline < aniline < p-toluidine (EWG lowers, EDG raises basicity).
(Q9.4(iv), gas phase): $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5NH_2 > NH_3$ (only $+I$ effect matters).

Pattern D: Conversions

Solved (Q9.5(i)): ethanoic acid $\xrightarrow{NH_3,\ \Delta}$ ethanamide $\xrightarrow{Br_2/NaOH}$ methanamine (Hoffmann, one C less).
(Q9.5(vi)): methanamine $\xrightarrow{HNO_2}$ methanol $\xrightarrow{PBr_3}$ CH_3Br $\xrightarrow{KCN}$ CH_3CN $\xrightarrow{LiAlH_4}$ ethanamine (one C more).

Pattern E: Identify A, B, C in a sequence

Solved (Q9.9(iv)): $C_6H_5NO_2 \xrightarrow{Fe/HCl} C_6H_5NH_2$ (A) $\xrightarrow{NaNO_2/HCl,\,273\,K} C_6H_5N_2{}^+Cl^-$ (B) $\xrightarrow{H_2O/H^+}$ C_6H_5OH (C).
(Q9.10): C (C_6H_7N) is aniline, so B is benzamide and A is benzoic acid.

Pattern F: Complete the reaction

Solved (Q9.11): (i) aniline + $CHCl_3$ + alc. KOH → phenyl isocyanide; (ii) diazonium + H_3PO_2 → benzene; (v) aniline + Br_2(aq) → 2,4,6-tribromoaniline; (vi) + $(CH_3CO)_2O$ → acetanilide.

Pattern A: Q9.1. Pattern B: Q9.2, 9.6. Pattern C: Q9.3, 9.4, 9.14. Pattern D: Q9.5, 9.8. Pattern E: Q9.9, 9.10. Pattern F: Q9.7, 9.11, 9.12, 9.13.

9D Exam tips

Common mistakes that lose marks

- Diazotisation needs 273–278 K; aliphatic diazonium salts decompose (give alcohols + N_2).
- To get mono-bromo or p-nitro aniline, protect $-NH_2$ by acetylation first.
- Hoffmann removes a carbon; nitrile reduction (via KCN) adds a carbon.
- Basicity order in water is not simply $3° > 2° > 1°$.

Question types: 1-mark: name reactions, basicity comparison; 2–3 marks: tests, conversions, reasons; 5 marks: sequences (A, B, C), reaction sets.

9E JEE extras

Useful results

- Hinsberg reagent: benzenesulphonyl chloride; 1° product dissolves in alkali (acidic N–H).
- Gattermann reaction: Cu/HCl or Cu/HBr instead of cuprous salts.
- Basicity in water (methyl series): $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$.

9F Pattern spotter

If you see...	Do this	See
Properties and tests		
Basic strength order	$+I$, solvation, steric; aryl amines weak	Ex C
1°/2°/3°	Hinsberg; carbylamine for 1°	Ex B
Aromatic 1° amine	Azo-dye test	Ex B
Synthesis		
One carbon less	Hoffmann bromamide	Ex D
One carbon more	Halide → nitrile → amine	Ex D
Replace NH_2 on benzene	Diazotise, then Sandmeyer etc.	Ex E
Sequence of reagents	Track each step's product	Ex E

Chapter 10 Biomolecules

10A Key facts and reactions

1. Reactions of glucose (open chain) and what they prove

HI, heat → *n*-hexane	six C in a straight chain
NH_2OH → oxime; HCN → cyanohydrin	carbonyl group
Br_2 water → gluconic acid	aldehyde (−CHO)
Acetic anhydride → pentaacetate	five −OH groups
HNO_3 → saccharic acid	a primary alcohol ($-CH_2OH$)

2. Cyclic structure: facts the open chain cannot explain

No Schiff's test and no $NaHSO_3$ addition; pentaacetate does not react with NH_2OH; existence of α and β anomers (mutarotation). Glucose forms a six-membered (pyranose) ring through C1–C5 hemiacetal.

10B Give-reasons bank / differences

Differences and definitions

Pair	First	Second
Starch vs cellulose	α-glucose; amylose + amylopectin	β-glucose; straight chains
Starch vs glycogen	Plant storage	Animal storage; more branched
Globular vs fibrous protein	Spherical, water soluble (insulin, albumin)	Fibre-like, insoluble (keratin, myosin)
Nucleoside vs nucleotide	Base + sugar	Base + sugar + phosphate
DNA vs RNA	Deoxyribose; A, G, C, T; double helix; heredity	Ribose; A, G, C, U; single strand; protein synthesis
Fat-soluble vs water-soluble vitamins	A, D, E, K (stored)	B group, C (must be taken regularly)
Essential vs non-essential amino acids	Must be in diet (valine, leucine)	Made by body (glycine, alanine)

10C Exercise-wise solution patterns

Exercises 10.1–10.25 (25 questions, 4 patterns)

Pattern A: Carbohydrates (definitions, classification, hydrolysis)

Solved (Q10.4): monosaccharides: ribose, 2-deoxyribose, galactose, fructose; disaccharides: maltose, lactose.
(Q10.7): sucrose $\rightarrow$ glucose + fructose; lactose $\rightarrow$ glucose + galactose.

Pattern B: Reactions of glucose

Solved (Q10.9): HI: *n*-hexane; bromine water: gluconic acid; HNO_3: saccharic acid.

Pattern C: Proteins and enzymes

Solved (Q10.14): α-helix is stabilised by intramolecular H-bonds between C=O and N–H of the chain.
(Q10.16): amino acids exist as zwitterions (H_3N^+–CHR–COO^-), so react with both acids and bases.

Pattern D: Vitamins and nucleic acids

Solved (Q10.19): fat-soluble (A, D, E, K) and water-soluble (B, C); vitamin K helps blood clotting.
(Q10.23): strands are complementary: A pairs with T, G with C, by H-bonds.

Pattern A: Q10.1–10.8. Pattern B: Q10.9, 10.10. Pattern C: Q10.11–10.18. Pattern D: Q10.19–10.25.

10D Exam tips

Common mistakes that lose marks

- Sucrose is non-reducing (glycosidic link between C1 of glucose and C2 of fructose); maltose and lactose are reducing.
- Denaturation destroys 2° and 3° structure, not the 1° sequence.
- D/L refers to configuration, not the sign of rotation.

Question types: mostly 1–3 marks: definitions, differences, glucose reactions, deficiency diseases (A: night blindness; C: scurvy; D: rickets; K: delayed clotting; B_{12}: pernicious anaemia).

10E JEE extras

Useful results

Invert sugar: sucrose (dextro) hydrolyses to a laevorotatory mixture. Anomers differ at C1; epimers (glucose and mannose) at C2. Isoelectric point: amino acid has no net charge. Peptide bond count in a polypeptide of n amino acids $= n - 1$.

10F Pattern spotter

If you see...	Do this	See
Carbohydrates		
Classify a sugar	Number of units on hydrolysis	Ex A
Reducing or not	Free hemiacetal OH?	10D
Glucose + reagent	Use the reaction table	Ex B
Proteins, vitamins, nucleic acids		
Structure levels	1° sequence; 2° H-bonds; 3° folding	Ex C
Vitamin role / deficiency	Memorise the list	Ex D
DNA vs RNA	Sugar, bases, strands, function	Ex D